

Physics Extended Essay – Detailed Structure Plan

Investigation on the frequencies of oscillations of a coupled
magnetic–mechanical oscillator

Working document (Edwin’s skeleton + INTERNATIONAL’s evaluation loop)

September 8, 2026

Storyline (keep this on your wall)

Two identical copper leaf springs with repelling magnets form two coupled oscillators. The in-phase mode (f_1) should not feel the magnets; the anti-phase mode (f_2) is stiffened by the magnetic coupling γ . Modelling the magnets as point dipoles gives

$$f_2^2 - f_1^2 = \frac{3\mu_0\mu_m^2}{\pi^3 m d^5},$$

i.e. a -5 power law. I measured f_1, f_2 by video tracking + FFT for $d = 21\text{--}60$ mm. The $\ln\text{--}\ln$ gradient was -3.75 ± 0.19 , f_1 drifted by 2.6 %, and points beyond 35 mm could not be resolved. The evaluation identifies four systematic effects (FFT resolution/leakage, amplitude-dependent non-linearity of the r^{-4} force, finite magnet size, non-identical springs). A modified method (normal-coordinate tracking, longer records, small release, time-domain two-tone fitting, non-linear least squares) plus an RK4 simulation of the full non-linear equations quantifies each effect and recovers the dipole law within its range of validity.

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1 Research question

RQ. *How does the equilibrium centre-to-centre separation d of the magnets in a coupled magnetic-mechanical oscillator affect its two normal-mode frequencies f_1 and f_2 , and does the coupling term $f_2^2 - f_1^2$ follow the inverse-fifth-power law predicted by the magnetic-dipole model?*

Why this wording: it names the independent variable (d), the dependent variables (f_1, f_2), the derived quantity that is actually tested ($f_2^2 - f_1^2$) and the theoretical claim (d^{-5}). Keep the title as it is; put the RQ in §1.3.

2 Mapping: Edwin’s structure → yours → INTERNATIONAL additions

Word budget is for a 4000-word essay (equations, tables, captions, appendices and references do not count). If you overrun, shrink 6.1 and 9 first; never shrink 10–13.

#	Edwin	Yours (section title)	Words	Extra from INTERNATIONAL
1	Introduction 1.1–1.3	1.1 Coupled magnetic-mechanical oscillator (done) · 1.2 Modes of oscillation and beats · 1.3 Motivation & RQ	350	motivation paragraph (their §1.4)
2	Qualitative prediction	2 Qualitative prediction	70	
3	Hypothesis	3 Hypothesis	50	
4	Variable table	4 Variables (table)	60	FFT settings listed as controlled variables
9	Theoretical model (Edwin puts it late)	5 Theoretical model (put it <i>before</i> methodology, as INTERNATIONAL does)	550	5.4 “Remarks on the model”: dimensional check, list of assumptions
5–7	Methodology, data collection, procedures	6 Methodology: 6.1 FFT background · 6.2 Set-up & apparatus · 6.3 Method of data collection · 6.4 Procedures	650	6.1 mirrors their §6.1 (DFT, resolution, leakage, windows)
8	Data	7 Raw data	200	half-range uncertainty and its justification
10	Processed data	8 Analysis: 8.1 processed table · 8.2 ln–ln graph · 8.3 Y vs d^{-5} and f_1 vs d · 8.4 comparison with theory	450	
11	Discussion & conclusion	9 Discussion (preliminary)	250	
12	Evaluation	10 Evaluation: systematic 10.1–10.5, random 10.6	550	each error with sign, size, remedy
–	–	11 Further methodology	200	their §11
–	–	12 Further data & discussion	350	their §12–13
–	–	13 Numerical simulation (RK4)	300	their §14
11.2, 13	Conclusion, extension	14 Conclusion & extensions	220	
A–E	Appendices	A–H (Section 9)	0	

3 Section-by-section content

1 Introduction

1.2 Modes of oscillation and beats. (≈ 150 words) Same ideas as Edwin’s 1.2 but for translational displacements x_1, x_2 of the magnets: in-phase mode (f_1 : magnets move together, separation constant, magnets “invisible”); anti-phase mode (f_2 : separation oscillates, magnetic force does work).

Any release is a superposition, hence beats with $f_{\text{avg}} = (f_1 + f_2)/2$ and $f_{\text{beat}} = (f_2 - f_1)/2$. Include the identity

$$4f_{\text{avg}}f_{\text{beat}} = f_2^2 - f_1^2, \quad (1)$$

which ties your test quantity to Edwin's average/beat language and explains why beats vanish at large d ($f_2 \rightarrow f_1$). Figure 2: your own sketch of the two modes (do not copy the IYPT slides). Figure 3: one of your own $x(t)$ traces showing beats.

1.3 Motivation & RQ. (≈ 150 words) Origin: IYPT 2023 problem 6 “Magnetic-mechanical oscillator” (cite the problem statement); analogy to contactless/magnetically coupled resonators; personal angle (coupling can be tuned continuously by moving the springs, unlike a fixed spring). State the RQ.

2 Qualitative prediction 3 Hypothesis

Repulsion $\propto d^{-4}$, so the stiffness added to the anti-phase mode, $\gamma = -dF/dd \propto d^{-5}$, collapses quickly with d : $f_2 \rightarrow f_1$ and the beat period $\rightarrow \infty$; f_1 stays constant.

Hypothesis: f_1 is independent of d ; $f_2^2 - f_1^2 \propto d^{-5}$, i.e. $\ln(f_2^2 - f_1^2)$ against $\ln d$ is a straight line of gradient -5 .

4 Variables (table, Edwin style)

Independent d (centre-to-centre, magnets mounted, measured at equilibrium).

Dependent f_1, f_2 (hence $f_2^2 - f_1^2$).

Controlled same spring pair; clamped length L ; clamp torque; same magnet pair and orientation; initial displacement (magnitude and which spring); release method; camera fps, resolution and distance; record length T ; FFT settings (window, N , zero-padding) – INTERNATIONAL lists FFT settings as controlled variables, do the same.

Uncontrolled air currents; temperature (E of copper); residual clamp slip.

5 Theoretical model (≈ 550 words)

5.1 Leaf spring as an oscillator. Cantilever tip stiffness $k = 3EI/L^3$; effective mass $m = M + \frac{33}{140}\mu L$ (Rayleigh) so $\omega_0^2 = k/m$ (derivation in Appendix A, or cite). One sentence: the Euler-Bernoulli continuum model of the IYPT slides reduces to this for a heavy tip mass; a finite-element check shows $< 1\%$ difference in $f_2^2 - f_1^2$ for $M/(\mu L) \geq 0.4$ (numbers in Section 10).

5.2 Magnetic force. Coaxial identical dipoles, like poles facing (from $\mathbf{F} = \nabla(\boldsymbol{\mu} \cdot \mathbf{B})$, Appendix B):

$$F(r) = \frac{3\mu_0\mu_m^2}{2\pi r^4} \quad (\text{repulsive}). \quad (2)$$

Equilibrium: springs bend outward by δ with $k\delta = F(d)$. You measure d directly, so the IYPT quintic for the equilibrium point is not needed – say so.

5.3 Linearised coupling and normal modes. $F(d + x_2 - x_1) \approx F(d) - \gamma(x_2 - x_1)$ with

$$\gamma = -F'(d) = \frac{6\mu_0\mu_m^2}{\pi d^5}. \quad (3)$$

Equations of motion

$$m\ddot{x}_1 = -kx_1 + \gamma(x_2 - x_1), \quad m\ddot{x}_2 = -kx_2 - \gamma(x_2 - x_1). \quad (4)$$

Normal coordinates $\eta = x_1 + x_2$, $\xi = x_1 - x_2$ (matrix form exactly like Edwin's eqs. 12–15):

$$\omega_1^2 = \frac{k}{m}, \quad \omega_2^2 = \frac{k + 2\gamma}{m} \quad \Rightarrow \quad \boxed{f_2^2 - f_1^2 = \frac{\gamma}{2\pi^2 m} = \frac{3\mu_0\mu_m^2}{\pi^3 m d^5}} \quad (5)$$

Linearised forms for graphs: (i) $\ln(f_2^2 - f_1^2) = \ln C - 5 \ln d$ with $C = 3\mu_0\mu_m^2/(\pi^3 m)$; (ii) $f_2^2 - f_1^2 = C d^{-5}$ (straight line through the origin against d^{-5} ; compare its gradient with C from independently measured μ_m, m).

5.4 Remarks (INTERNATIONAL §5.3 style). Dimensional check $[\mu_0\mu_m^2/(md^5)] = 1/s^2$. List the assumptions you will revisit: (a) point dipoles, (b) linearised force (small amplitude), (c) identical springs, (d) lumped mass, (e) no damping, (f) magnets stay coaxial, (g) the FFT returns the true frequencies.

6 Methodology (≈650 words)

6.1 FFT background – INTERNATIONAL §6.1 adapted to *frequency* measurement; the physics you need is in Section 4.

6.2 Set-up & apparatus: labelled photo (Edwin Fig. 3), apparatus list, magnet specification (grade, $D \times t$, mass), spring dimensions (L, b, h), camera (model, fps, resolution), tracking software.

6.3 Method of data collection – narrate like Edwin §6: Method 1 (read beat and average periods from $x(t)$) failed at large d because the envelope disappears when $f_2 - f_1$ is small; Method 2 (FFT of $x(t)$) adopted. Show one $x(t)$ and its FFT with the two peaks labelled (Edwin Figs. 4–6).

6.4 Procedures:

6.4.1 spring stiffness k : static load–deflection, plot F vs x , LINEST (Edwin §7.1);

6.4.2 magnet moment μ_m : either datasheet B_r ($\mu_m = B_r V/\mu_0$) or, better, one magnet on an electronic balance, the other brought coaxially above it at measured r , $F = \Delta m g$, fit $F \propto r^{-4}$ (doubles as an independent check of the r^{-4} law);

6.4.3 masses $M, \mu L$;

6.4.4 f_1, f_2 vs d : set d , displace one spring by $A = 10$ mm, release, record T s at f_s fps, track, FFT, read the two highest peaks, 3–5 trials per d .

7 Raw data (≈200 words)

Table 1 basic quantities ($L, b, h, M, \mu, k \pm \Delta k, \mu_m \pm \Delta\mu_m, m$). Table 2 F vs x for k . Tables 3/4 f_1 and f_2 per trial with mean and half-range (justify half-range vs standard deviation as INTERNATIONAL §8.2 does: $n = 3-5$ is too small for SD). **State T, f_s, N and the bin width $1/T$ in the caption** and add a column “resolved? ($f_2 - f_1 \geq 2/T$)”. Rows $d = 40-60$ mm: “peaks unresolved” – this is a result, not a gap.

8 Analysis (≈450 words)

8.1 Processed table: $d, \ln d \pm \Delta d/d, Y = f_2^2 - f_1^2, \Delta Y = 2(f_1 \Delta f_1 + f_2 \Delta f_2), \ln Y \pm \Delta Y/Y, d^{-5} \pm 5(\Delta d/d) d^{-5}$. One sample calculation (Edwin style).

8.2 Graph 1: $\ln Y$ vs $\ln d$ with error bars, best- and worst-fit lines (or LINEST if the bars are too small). Report the gradient -3.75 ± 0.19 (unweighted) and note the weighted value -3.48 ± 0.24 – the disagreement itself exposes the $\ln$ -transform problem.

8.3 Graph 2: Y vs d^{-5} (theory: straight line through the origin, gradient C). Graph 3: f_1 vs d (theory: horizontal) – you will see the 2.6% drift.

8.4 Comparison table (Edwin Tables 13/14): theoretical gradient -5 (exact) vs experimental; theoretical C from μ_m, m vs the experimental gradient of Graph 2; percentage differences.

9 Discussion – preliminary (≈250 words)

Trend confirmed (Y falls steeply, $f_2 \rightarrow f_1$), but: exponent $-3.75 \neq -5$; f_1 not constant; points beyond 35 mm unresolved; residuals systematic (curvature) rather than random. Frame the four candidate causes and hand over to §10.

10 Evaluation (≈ 550 words)

One sub-section per effect, each with *sign*, *size*, *remedy*.

10.1 FFT resolution & leakage (methodological/procedural). Two tones closer than $\sim 1/T$ (rectangular) or $\sim 2/T$ (Hann) merge or “pull” each other; your $f_2 - f_1 = 0.039$ Hz at 35 mm needs $T \geq 26$ –51 s, and 0.009 Hz at 40 mm needs $T \geq 110$ –220 s. Show the synthetic-spectrum figure built from *your* f_1, f_2 (Fig. 1). Sign: biases the large- d end and flattens the gradient. Remedy $\rightarrow$ §11.

10.2 Non-linear magnetic force (assumption b). Expand $F(d - \xi)$:

$$\ddot{\xi} + \omega_2^2 \xi + \alpha \xi^2 + \beta \xi^3 = 0, \quad \alpha = \frac{5}{2} \frac{g \omega_2^2}{d}, \quad \beta = \frac{5}{d^2} g \omega_2^2, \quad g = \frac{f_2^2 - f_1^2}{f_2^2}. \quad (6)$$

Landau–Lifshitz amplitude shift (Appendix D):

$$\frac{\Delta \omega}{\omega_2} = \left(\frac{a}{d}\right)^2 \left(\frac{15g}{8} - \frac{125g^2}{48}\right), \quad a = \text{amplitude of } \xi. \quad (7)$$

With your $A = 10$ mm release: +7.6 % at 21 mm (RK4 exact: +9.8 %), +1.2 % at 35 mm (Fig. 2). Sign: f_2 overestimated at small d , gradient *steeper* (−5.16 in simulation, Fig. 3) – a large error but not the cause of the flattening. Remedy: $A \leq 3$ mm, analyse late (small-amplitude) segments, extrapolate f_2 to zero amplitude, RK4 correction (§13).

10.3 Finite magnet size (assumption a). Replace point dipoles by two uniformly “charged” discs per magnet (Gilbert model, Appendix E). For d only 2–4 $\times$ the magnet diameter the near field is softer than d^{-5} : apparent gradient over 21–35 mm is −4.9 ($D = 6$ mm), −4.5 ($D = 12$ mm), −3.75 ($D = 20$ mm) (Fig. 4; run `finite_size_magnets.py` with *your* D, t). Sign: flattening – the leading candidate if your magnets are ≥ 12 mm. Remedy: exact cylindrical-magnet force in the model, or larger d (which collides with 10.1 – discuss the trade-off explicitly).

10.4 Non-identical springs (assumption c) – analogue of Edwin’s 12.2/Appendix D. With uncoupled frequencies $f_a \neq f_b$ (Appendix C):

$$\omega_{1,2}^2 = \bar{\omega}^2 + \kappa \mp \sqrt{\Delta^2 + \kappa^2}, \quad \kappa = \frac{\gamma}{m}, \quad \Delta = \frac{\omega_b^2 - \omega_a^2}{2} \implies (f_2^2 - f_1^2)^2 = (f_b^2 - f_a^2)^2 + (Cd^{-5})^2. \quad (8)$$

Consequences: f_1 rises as d shrinks (observed 0.802 $\rightarrow$ 0.823 Hz; the model with $f_a = 0.793$, $f_b = 0.849$ Hz gives 0.798 $\rightarrow$ 0.819 Hz) and Y saturates at $|f_b^2 - f_a^2|$ instead of $\rightarrow 0$ – flattening at large d . Remedy: measure f_a, f_b with the other magnet removed; fit the detuned formula.

10.5 Damping (assumption e). Fit the envelope, get τ ; frequency shift $\approx \frac{1}{2}(1/4\pi f\tau)^2$ – negligible for $\tau \sim 20$ –30 s – but the Lorentzian linewidth $\approx 1/(\pi\tau)$ (≈ 0.01 –0.02 Hz) sets a floor on resolvability that no record length removes (feeds 10.1). Lumped-mass assumption (d): $< 1\%$ (Appendix A) – dismiss in one sentence, with the number.

10.6 Random errors: d (± 0.5 mm, $\Delta(\ln d) = 2\%$ at 21 mm), tracking noise (sub-pixel), frame-timing jitter, release repeatability (half-range across trials), bin-centring/scalloping $\pm 1/(2T)$ unless interpolated.

11 Further methodology (≈ 200 words)

Numbered modified procedure, INTERNATIONAL §11.2 style:

1. Measure f_a, f_b of each spring alone (other magnet removed).
2. Track **both** magnets; form $\eta = x_1 + x_2$ and $\xi = x_1 - x_2$ *before* the FFT. Each normal coordinate contains essentially **one** frequency, so the two-peak resolution problem disappears – f_1 from η , f_2 from ξ , even at 45–60 mm. (Physically motivated, free, and the single best improvement.)
3. Record ≥ 90 s at 60 fps (or 30 fps), tripod, scale bar in frame; release $A \leq 3$ mm; also one run at $A = 10$ mm per d for the amplitude study.

4. Re-analyse every record with rectangular / Hann / Blackman windows + $16\times$ zero-padding + parabolic peak interpolation (INTERNATIONAL Table 10 style comparison), **and** with a time-domain two-damped-sinusoid least-squares fit (beats the $1/T$ limit; gives a statistical Δf).
5. Fit models to (d, f_1, f_2) by weighted non-linear least squares: (a) dipole, $n = 5$; (b) free exponent n ; (c) finite-size disc model; (d) detuned model. Compare residuals/ χ^2 .

`analyse_tracks.py` performs steps 4–5 for a folder of Tracker exports.

12 Further data & discussion (≈ 350 words)

Table: gradient $\pm\Delta$ for each method (rect / Hann / Blackman / two-tone fit / normal-coordinate FFT) – like INTERNATIONAL Figs. 26–27. Graph: f_2 vs amplitude (segments) with extrapolation to zero amplitude. Graph: $\ln Y$ vs $\ln d$ for old vs new data with both theory curves (point dipole and finite size). Discuss which correction moved the gradient toward -5 and by how much; be explicit if a residual discrepancy remains and what it implies (e.g. “the dipole law is valid only for $d \gtrsim 4D$ ”).

13 Numerical simulation (≈ 300 words)

RK4 integration of the **full** equations

$$m\ddot{x}_1 = -kx_1 - [F(d+x_2-x_1) - F(d)], \quad m\ddot{x}_2 = -kx_2 + [F(d+x_2-x_1) - F(d)], \quad F = C_F r^{-4}, \quad (9)$$

(optionally with the finite-size F) using your measured k, m, C_F . Two uses: (i) predict $f_2(A, d) \rightarrow$ the amplitude correction used in §12; (ii) generate synthetic $x(t)$ at your fps and T and push them through the *same* FFT pipeline $\rightarrow$ isolates the bias caused purely by the analysis (in my test at $T = 40$ s, window peak-picking gave gradients -3.3 to -3.5 while the two-tone fit recovered the true value). Code $\rightarrow$ Appendix G.

14 Conclusion & extensions (≈ 220 words)

Answer the RQ in three sentences (f_1 constant within $\dots$; $f_2^2 - f_1^2$ falls as d^{-n} with $n = \dots \pm \dots$; the dipole law holds for $d \geq \dots$). Name the dominant systematic error and what fixed it. Extensions: vary μ_m (magnet grade) at fixed d to test the μ_m^2 dependence; attractive orientation (softening $\rightarrow$ instability at a critical d); three springs; eddy-current damping with a copper plate.

4 FFT error analysis – what to write

INTERNATIONAL needed **amplitudes** (SNR), so its corrections were energy-correction factors and baseline-noise subtraction. You need **frequencies**, so the relevant FFT physics is different – say this explicitly.

1. **Resolution.** $X_k = \sum_{n=0}^{N-1} x_n e^{-2\pi i k n / N}$; bin spacing $\Delta f_{\text{bin}} = f_s / N = 1/T$. Two tones are separable only if $|f_2 - f_1| \gtrsim 1/T$ (rectangular) or $\gtrsim 2/T$ (Hann; main lobe twice as wide). From your data:

d/mm	$f_2 - f_1/\text{Hz}$	bins at $T = 20$ s	T_{min} rect/s	T_{min} Hann/s
21	0.240	4.8	4	8
25	0.141	2.8	7	14
28	0.097	1.9	10	21
31	0.074	1.5	14	27
35	0.039	0.8	26	51
40	0.009	0.2	111	222

2. **Leakage & windows.** Rectangular: narrowest main lobe (best for two close peaks of similar height) but -13 dB side lobes; Hann: 4-bin main lobe, -31 dB; Blackman: 6-bin, -58 dB.

Because both your peaks have similar amplitude and there is no strong interferer, the rectangular (or Hann) window is the right choice – the opposite of the usual “Hann by default” advice, and unlike INTERNATIONAL’s case. Amplitude correction factors are **not** needed because you only read peak positions.

3. **Peak pulling.** When main lobes overlap the two maxima are pushed apart or merge; in my synthetic test ($T = 20$ s) the errors were $\pm 2\text{--}7$ mHz at 21–31 mm, ± 10 mHz at 35 mm and total failure at 40 mm – quote your own numbers from `ee_simulations.py` Part A with your real T and f_s .
4. **Scalloping / bin-centring.** The true frequency lies anywhere within $\pm \frac{1}{2}$ bin, i.e. $\pm 1/(2T)$, unless you zero-pad and interpolate (parabola through the three bins around the peak). Zero-padding interpolates the spectrum, it does *not* add resolution – make that distinction.
5. **Damping linewidth.** $e^{-t/\tau} \cos(2\pi f t)$ has $\text{FWHM} \approx 1/(\pi\tau)$; with $\tau \approx 25$ s that is 0.013 Hz – comparable to $f_2 - f_1$ at 40 mm, so beyond ~ 35 mm the modes cannot be separated by any spectral method applied to a single coordinate $\rightarrow$ justification for normal-coordinate tracking and time-domain fitting.
6. **Sampling.** $f_s = \text{fps}$ (30/60 Hz) $\gg 2f$, no aliasing; but video frame timing jitters and frames drop $\rightarrow$ resample to a uniform grid before the FFT (the script reports the jitter).
7. **Propagation into your graph.** $\Delta(\ln Y) = 2(f_1\Delta f_1 + f_2\Delta f_2)/Y$ grows from 0.04 to 1.1 across your range for $\Delta f = 5$ mHz, so error bars must be drawn in ln-space and the fit weighted; otherwise the noisy large- d points dominate the gradient.

5 “Non-linearisation” improvements – three levels (use all three, in order)

Level 1 – analysis (cheap, do it now).

Replace “take logs then LINEST” with a weighted non-linear least-squares fit of $Y = Cd^{-n}$ (and $Y = Cd^{-5}$ with n fixed) to the un-transformed data. Advantages to state: no point is discarded because $\ln(0)$ or $\ln(\text{negative})$ fails; heteroscedastic errors are handled by the weights; C and n come with proper uncertainties. With your 6 points: $n = 3.5 \pm 0.2$ (vs $\ln\text{--}\ln$ -3.75) – report both and explain the difference.

Level 2 – physics (the systematic-error story).

The model linearised $F(r)$ about d . Keep the full r^{-4} (or finite-size) force: perturbation formula (Appendix D) in the body, RK4 (§13) for exact numbers, amplitude-segment analysis + extrapolation to $A \rightarrow 0$ for the data. Tabulate the size of the correction at each d .

Level 3 – model (finite size + detuning).

Fit models (c) and (d) of §11 step 5. Present the comparison of residual patterns as the decisive evidence for which assumption failed.

6 Lab / data checklist (you have lab access and the raw tracks)

- ☐ Magnet dimensions D, t , mass M , grade (B_r) – needed for §10.3 and the theoretical C .
- ☐ Spring L, b, h, μ (weigh a known length), k by static loading; f_a, f_b of each spring alone.
- ☐ Balance measurement of $F(r)$ for 4–6 values of r (checks r^{-4} and gives μ_m).
- ☐ Re-run $d = 21 \dots 60$ mm with both magnets tracked, $T \geq 90$ s, $A = 3$ mm (and one $A = 10$ mm run each). Name files `d21_trial1.csv` etc.
- ☐ `python analyse_tracks.py "tracks/*.csv" --out results` $\rightarrow$ per-file diagnostic figure ($x(t)$ + two-tone fit, three-window FFT, f_2 vs amplitude), `summary.csv`, and the $\ln\text{--}\ln$ gradient per method. Then form η, ξ from the two tracked magnets and run the same script on them.
- ☐ Sanity check of magnitudes: $\gamma = 2\pi^2 m Y$ from your data vs $\gamma = 6\mu_0 \mu_m^2 / (\pi d^5)$ from μ_m . If they differ by more than a factor ~ 2 , revisit the coaxial-dipole assumption or the definition of d before writing §8.4.

- Re-derive the $f_2 - f_1$ requirement with your real T : if your original records were $\sim 20\text{--}40\text{ s}$, say in §7 that 35 mm was marginal and 40 mm unresolvable.

7 Figures and tables (final list)

Figures: 1 set-up sketch (done) · 2 mode sketch · 3 set-up photo · 4 $x(t)$ with beats (small d) · 5 $x(t)$ with no visible envelope (large d) · 6 FFT with two peaks · 7 F vs $x(k)$ · 8 F vs r on the balance (μ_m) · 9 $\ln Y$ vs $\ln d$ (Graph 1) · 10 Y vs d^{-5} (Graph 2) · 11 f_1 vs d (Graph 3) · 12 synthetic-spectrum resolution figure · 13 f_2 vs amplitude · 14 gradient by method (further data) · 15 old vs new data with dipole and finite-size theory · 16 RK4 vs linearised prediction.

Tables: 1 constants · 2 k data · 3/4 f_1, f_2 raw · 5 processed · 6 theory vs experiment (gradient, C) · 7 systematic-error summary (effect, sign, size, remedy) · 8 further raw data (appendix) · 9 gradient/ C per method and per model · 10 simulated vs measured f_2 shift.

8 Supporting figures already generated

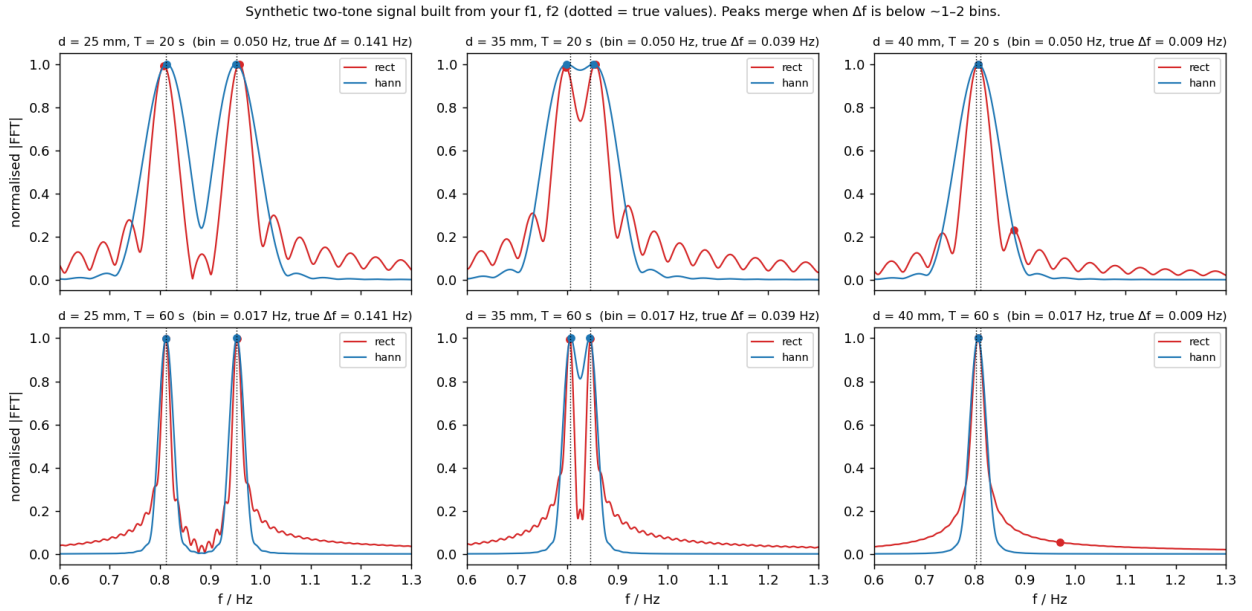


Figure 1: Synthetic two-tone spectra built from your f_1, f_2 (dotted lines = true values) for $T = 20\text{ s}$ and 60 s . Peaks merge when $f_2 - f_1$ falls below $\sim 1\text{--}2$ bins. Use in §10.1.

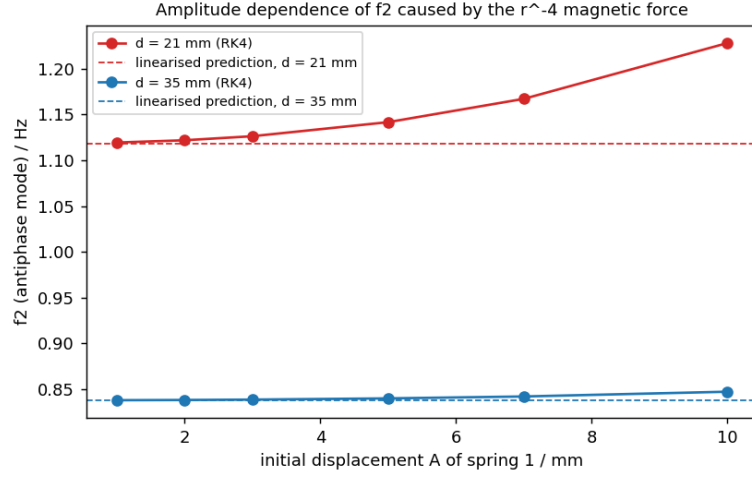


Figure 2: RK4 simulation: amplitude dependence of f_2 caused by the r^{-4} force, compared with the linearised prediction. *Use in §10.2 / §13.*

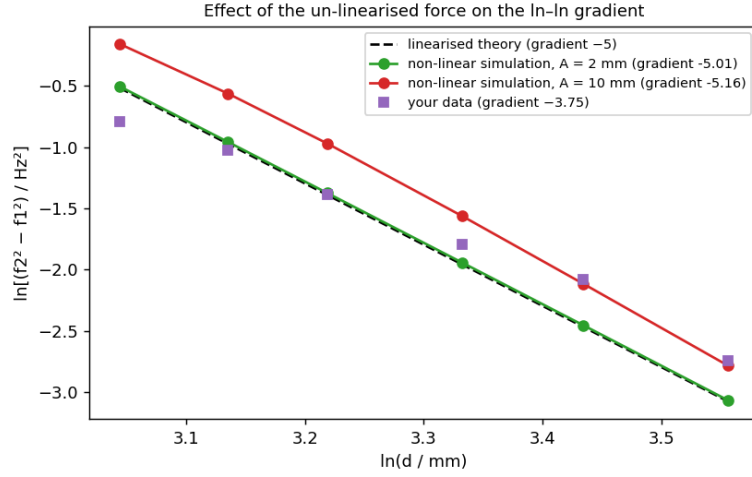


Figure 3: Effect of the un-linearised force on the ln-ln gradient: linearised theory (-5), non-linear simulation at $A = 2$ and 10 mm, and your data. *Use in §10.2.*

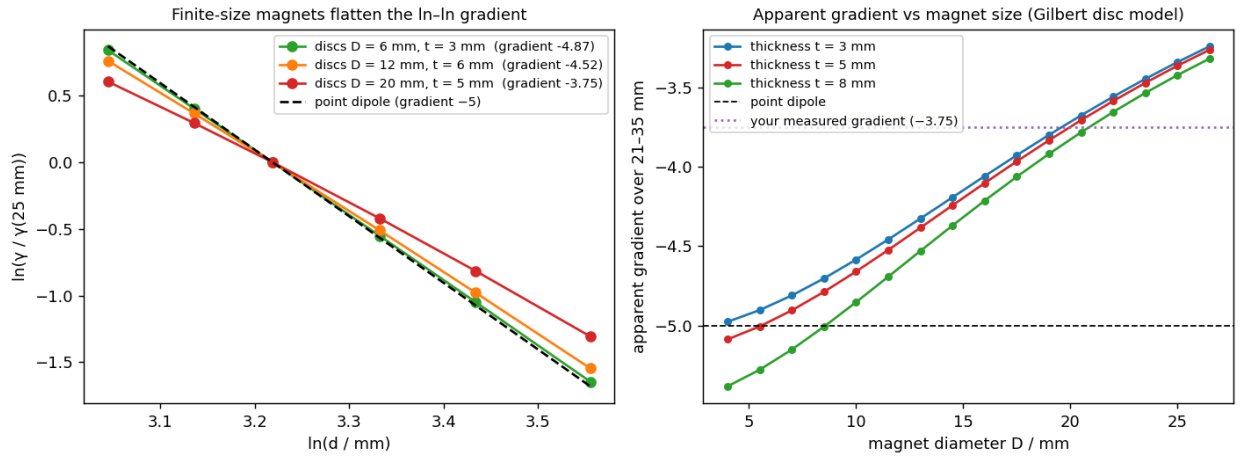


Figure 4: Finite-size (charged-disc) magnet model: apparent ln-ln gradient over $21\text{--}35$ mm versus magnet diameter and thickness. *Use in §10.3.*

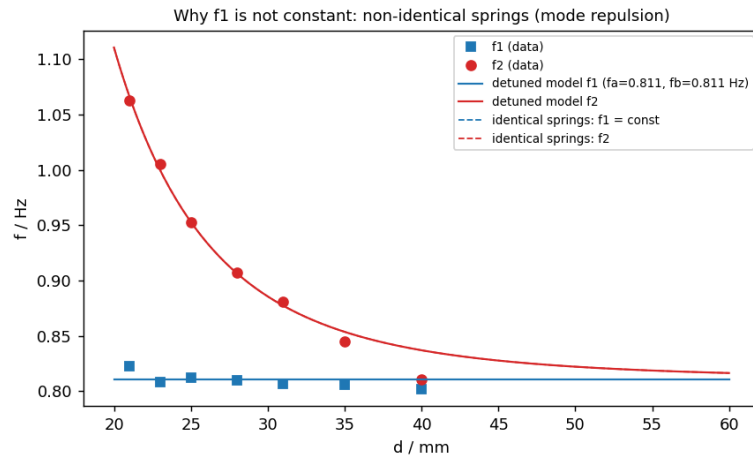


Figure 5: Why f_1 is not constant: detuned (non-identical springs) model versus the identical-spring model. *Use in §10.4.*

9 Appendices to write

- A** Cantilever stiffness $3EI/L^3$, effective mass $\frac{33}{140}\mu L$, and FEM check of lumped vs beam model ($< 1\%$).
- B** Dipole field $\rightarrow$ force between coaxial dipoles $F = 3\mu_0\mu_m^2/(2\pi r^4) \rightarrow \gamma = 6\mu_0\mu_m^2/(\pi d^5)$.
- C** Detuned normal modes $\omega_{1,2}^2 = \bar{\omega}^2 + \kappa \mp \sqrt{\Delta^2 + \kappa^2}$ and the d^{-5} limit.
- D** Series expansion of $F(d - \xi)$, α and β , Landau-Lifshitz amplitude shift $(a/d)^2(15g/8 - 125g^2/48)$, comparison with RK4.
- E** Finite-size (charged-disc) magnet force and apparent-gradient table.
- F** Uncertainty propagation (ΔY , $\Delta \ln Y$, Δd^{-5} , LINEST worst lines).
- G** Python: `analyse_tracks.py` (FFT windows, two-tone fit, segments), `ee_simulations.py` (RK4, synthetic FFT), `finite_size_magnets.py`.
- H** Further raw data tables (all windows/methods), as INTERNATIONAL Appendix A.

10 Ready-made numbers from my checks

Replace with your own values once you have real inputs

Assumed in the scripts: f_s , τ , magnet size and mass – marked # ASSUMED.

- ln–ln fit, 6 points: gradient -3.75 ± 0.19 , intercept 10.68 (matches your chart); weighted ($\Delta f = 5$ mHz): -3.48 ± 0.24 . Free power law on raw Y : $n = 3.5 \pm 0.2$.
- Non-linear force, $A = 10$ mm release: $f_2 + 9.8\%$ (21 mm), $+1.1\%$ (35 mm); gradient $-5.00 \rightarrow -5.16$. $A = 2$ mm: $+0.3\%$ / $+0.03\%$.
- Finite-size discs, apparent gradient over 21–35 mm: $D = 6$ mm -4.87 · 10 mm -4.6 · 12 mm -4.5 · 15 mm -4.2 · 20 mm -3.75 · 25 mm -3.4 ($t = 3$ –6 mm changes these by < 0.1).
- Lumped vs beam: Y differs by 0.03% ($M/\mu L = 3.3$), 0.4% (0.8), 1% (0.4).
- Detuned model fitted with $n = 5$: $f_a = 0.793$ Hz, $f_b = 0.849$ Hz reproduces the f_1 drift $0.798 \rightarrow 0.819$ Hz.
- Synthetic tracks at $T = 40$ s: window peak-picking gradients -3.3 to -3.5 ; two-tone fit -4.9 (true -4.9) $\rightarrow$ the analysis method alone can shift the gradient by more than 1.

11 Pitfalls to avoid (examiner’s eye)

- Do not present the 40 mm point as data; present the unresolved range as a measured limit of the method.
- Do not claim “ f_1 constant” when it drifts 2.6%; explain it (10.4).
- Do not copy IYPT slide figures or equations without derivation; cite the IYPT problem and derive the lumped model yourself.
- Keep all corrections quantitative (sign and size) – hand-waving loses evaluation marks.
- Every graph: error bars, best/worst lines or LINEST uncertainty, units, gradient with uncertainty, theory line overlaid where possible.
- Word count: equations, tables, captions, appendices excluded – move derivations there ruthlessly.